Figure 2 Gate

Fitness and Gene–Gene Interactions: What the Lower-Order Terms Explain, What the Transformer Recovers, and the Panels Proposed for the Main Text

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1 The decision ✗

Figure 2 of the manuscript is to carry fitness and gene–gene interactions, with gene essentiality moved to its own figure. Every arm here is scored on the pinned held-out triples of the Kuzmin 2018 and 2020 trigenic screens on the 030 build: arm R, the 010 random split over records (301,386 train, 37,673 validation, 37,673 test triples), and arm Q, the query-pair-disjoint split (301,236, 37,705, 37,791), in which no held-out triple's query pair appears in any training record. Nothing here is approved. The sheets are mockups of what the result files support, and the symbol after each panel letter says only whether the values behind it are on file.

Where the full record is. `025-additive-baselines` holds the gene-identity ladder on both arms, the arms and their missing control, the prespecified decision table and the first arm Q test score of a transformer checkpoint. `025-s3-closure` holds the closure: the published trigenic score reproduced from the stored lower-order records, and what that identity says about training on the closure pool. `025-mmli-campaign` and `030-scaling-study` hold the transformer campaigns on the 025 and 030 builds. The lower-order ladder of this document is the script `lower_order_baselines_030.py` and its note.

The question the figure answers, and the one it must not dodge. The joint model of experiment 030 trains on every single, every closure double and every training triple of the S3 pool, with a fitness head beside the interaction head. For a held-out triple the published identity

$$\tau_{ijk} = f_{ijk} - f_{ij}\, f_k - \varepsilon_{ik} - \varepsilon_{jk} \tag{1}$$

has every term but f_{ijk} in the training pool: the query strain's double fitness f_{ij}, the array single f_k and the two digenic control scores. So the model is handed the triple's null expectation, and the question a reviewer will ask is whether its score is that arithmetic. The gene-identity ladder of 010 and 025 cannot answer it, because a one-hot ridge over the three genes never sees a measurement. The lower-order ladder can: every rung reads only those terms, under the trainer's own label policy, and on arm Q the terms of any held-out query pair are masked (74,404 rows lose f_{ij} and 68,920 lose ε_{ij}), exactly the doubles the strict closure rule removed from the model's training pool.

Apples to apples, stated precisely. The information set is the same on both sides: the lower-order rungs and the transformer see the identical training records, and neither sees the held-out triple's own fitness or interaction, since the split is by record and a 030 record holds all of a genotype's entries. What differs is the access mode, and the sheets name it as a regime. A lower-order rung is a lookup plus a function: the measured terms are inputs at scoring time, which is transductive over the closure. The transformer as trained reads gene identity, graphs and sequence embeddings at scoring time and must have stored the lower-order records in its weights. So the ladder is not a claim that boosted

trees are the better model of a cell. It is the measurement of how much of a held-out trigenic score is already in the lower-order records, and the model's distance from that number is the thing the figure has to show. A transformer that reads the same terms as inputs is the arm that makes the two regimes directly comparable, and it is the first entry of the registry (6).

CGT is a fitness predictor, and that is the other fair reading. The joint model predicts a held-out triple's fitness as well as its score, and on arm R its fitness head reads 0.949 against 0.927 for the ridge over the terms. Equation 1 then gives two rungs that use CGT as what it is. C1 takes the model's predicted f_{ijk} and subtracts the measured null, $f_{ij}f_k + \varepsilon_{ik} + \varepsilon_{jk}$, the same terms the lower-order rungs read, so the model supplies only the one term the closure lacks; it is the comparison in which the model and the trees receive the same information in the same form. C2 derives τ from the model's own predicted fitness of the triple, the doubles and the singles, with no measured term, which tests whether its fitness predictions are consistent enough for the residual. Neither is on file: both need the fitness head's per-triple predictions, which E3 writes beside the interaction head's. If C1 reads above the trees, the model's fitness head is doing the work the ladder measures and the figure says so; if it does not, the fitness head is no more precise on f_{ijk} than a ridge over the terms, which Figure 2f already suggests on arm Q.

Three statements are on file. First, the free arithmetic is large. The zero-parameter null that predicts τ as minus the two digenic control scores reads 0.577 on arm R and 0.566 on arm Q (test Pearson), boosted trees over the thirteen terms and four products 0.743 and 0.700, against the gene-identity ridge at 0.401 and 0.185; the paired gain of the trees over the ridge is +0.342 [+0.327, +0.359] on R and +0.515 [+0.501, +0.529] on Q over 2,000 resamples of the test triples. Second, the transformer does not collect it. The composite-embedding model with the fitness head and the dataset token reads 0.562 on arm R (validation, mean of epochs 155 to 164, seed 0) and 0.399 on arm Q (epochs 36 to 45), under the zero-parameter null on both arms, and on Q within a screen it reads close to the screen-mean floor. Third, the fitness side is arithmetic. A ridge over the same terms predicts a held-out triple's fitness at 0.927 on R and 0.734 on Q, and the model's fitness head reads 0.949 and 0.722 (validation) on the same triples, so the fitness panel of the figure is a panel about the closure, not about the model.

Every pooled number carries a floor. The two screens differ in mean τ, and predicting only the training mean of the triple's screen reads 0.368 on R and 0.378 on Q. The dataset token learns that offset in the first epoch. Every comparison on the sheets is therefore drawn with the floor visible, and the performance table (7) carries the within-screen Pearson, both series centered by screen, beside the pooled one.

The claims the main-text figure would make. First, that the trigenic score is a residual over lower-order measurements, and the ontology stores those measurements beside it, which is what makes the closure recomputable and the free arithmetic measurable: on file. Second, that a model reading gene identity and sequence recovers part of a held-out triple's score on a random split and less on a query-pair-disjoint split: on file for one seed, validation only. Third, that the model beats what the lower-order records already say: not on file, and on the current reading false on both arms. Fourth, that the figure can show the comparators of the earlier draft (DCell, DANGO, FBA) on the same data: not on file; their results are on other builds (3). The figure enters the paper only when the third claim is settled either way by the registry's first two entries, and with the regime named in the panel.

2 The mockups ✗

Result files are under `experiments/030-solid-growth-multi/results/` unless a path is given. Each caption gives, per panel, what is plotted, the statistic, the sample size and the status. The statistic is the Pearson correlation on the pinned test triples of the arm unless the caption says otherwise; a transformer row is a validation number, the mean over the last ten epochs synced to the run registry, because no 030 checkpoint has a test evaluation yet. Regime is what a method reads when it scores a held-out triple: *identity*, the three genes and nothing measured (the gene-identity ridge, and the transformer as trained, which sees the lower-order records only through its weights); *lower-order terms*, the measured singles, doubles and digenic scores of the same strains, looked up at scoring time (a transductive read over the closure); *screen only*, the training mean of the triple's screen. Light fill is a test score, the darker line color a validation score of a model with no test evaluation.

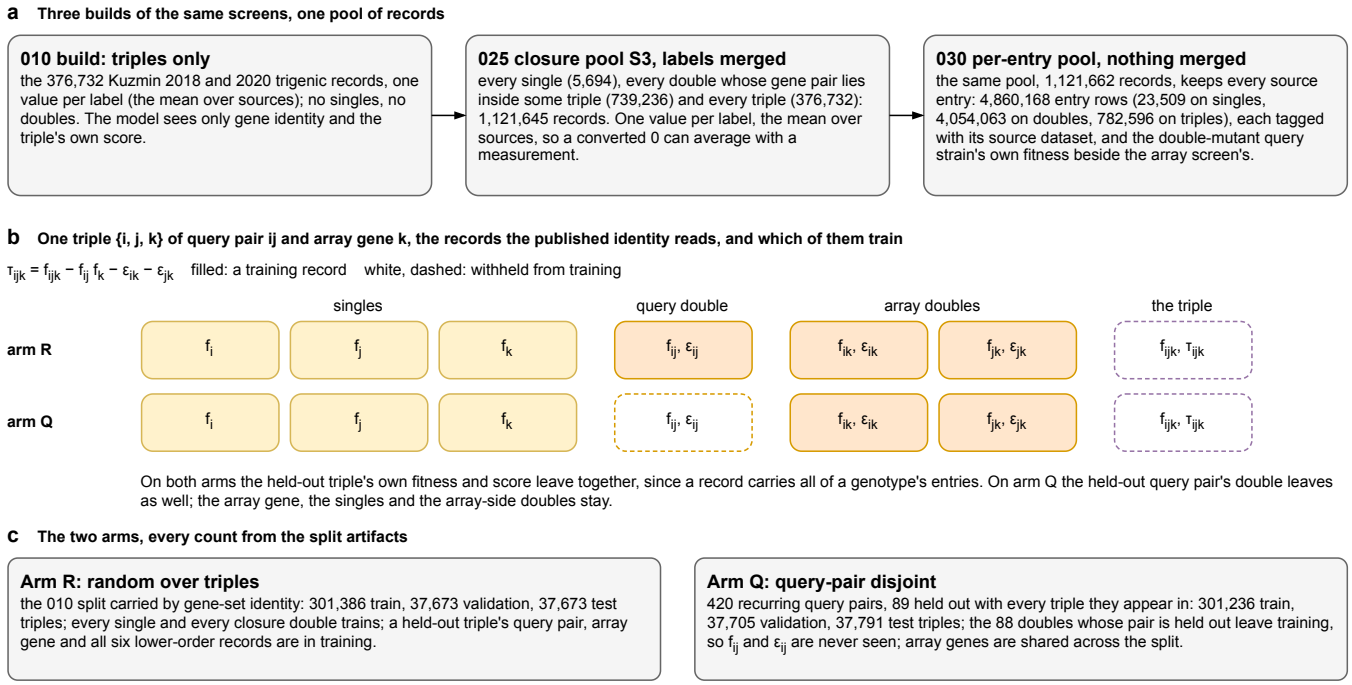


Figure 1: The datasets behind the ladder: three builds of the same screens, the records one triple's identity reads, and the two arms. a, The 010 build holds the 376,732 Kuzmin trigenic records only; the 025 closure pool adds every single and every double whose pair lies inside some triple, with one value per label as the mean over sources; the 030 pool keeps every source entry of the same records as its own row with a dataset token, and adds the double-mutant query strain's own fitness beside the array screen's. **b**, One triple of query pair ij and array gene k and the seven records the identity of equation 1 reads; a filled chip is a training record, a white dashed chip is withheld. On both arms the triple's own fitness and score leave together, since a record carries all of a genotype's entries; on arm Q the held-out query pair's double leaves as well, and the singles, the array gene and the array-side doubles stay. **c**, The two arms with their counts from the split artifacts. Generated by `notes/assets/drawio/030-datasets-splits.gen.py` from `subset_definitions_030_summary.json`, `subset_definitions_030_q_summary.json` and `transfer_010_tmi_splits_030_summary.json`; composed from the split scripts' definitions, not redrawn from a published figure. ●

The sheet proposed for the main text. Figure 2, built the way the manuscript's Figure 1 is built: one dense page, every panel from a committed result file, color the input family a method reads at scoring time. The symbols after each panel letter are its section status (not reviewed) and figure status, with the key at the sheet's foot. Panels a and l are compositions with no data behind them yet; j and k are partial; the rest are on file.

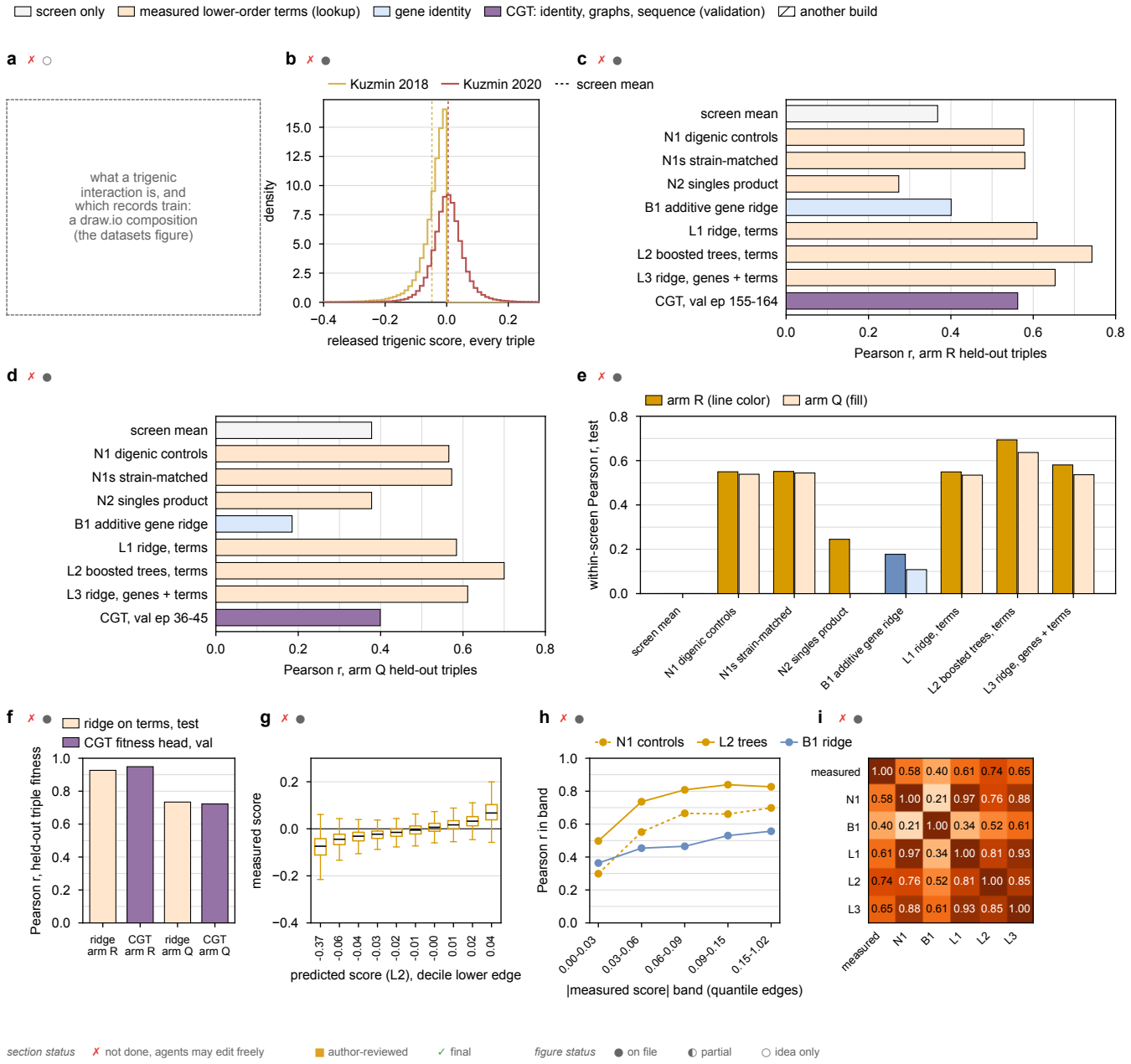


Figure 2: The sheet proposed for the main text: what a held-out trigenic score already is in the lower-order records, what each method recovers of it, and the fitness side. Color is the input family: gray the screen alone, orange the measured lower-order terms of the same strains (a lookup, transductive over the closure), blue gene identity alone, purple CGT reading identity, the gene graphs and sequence embeddings; light fill is a test score and the darker line color a validation score of a model with no test evaluation. Written by `figure2_gate_sheet.py`, GilaHyper, 2026-10-12. The panel-by-panel reading is under the figure.

What each panel of Figure 2 reports. **a**, What a trigenic interaction is and which of its records train on each arm, the composition of Figure 1. ✗○ **b**, The released score of every triple by screen: the two screens differ in mean (dashed), which is the screen-offset floor every pooled Pearson carries (0.368 on arm R test). ✗● **c**, The interaction ladder on arm R, test Pearson on 37,673 held-out triples: the screen mean, the zero-parameter digenic-controls null (0.577), its strain-matched form, the singles-product null, the additive gene ridge (0.401), ridge and boosted trees over the lower-order terms (0.609, 0.743), ridge over genes and terms, and CGT's validation mean over epochs 155 to 164 (0.562). ✗● **d**, The same ladder on arm Q over 37,791 held-out triples, where the held-out query pair's double is withheld from every rung as from the model (0.566, 0.185, 0.700; CGT 0.399 over epochs 36 to 45). ✗● **e**, Every rung's within-screen Pearson on test, both series centered by screen, arm R in the line color and arm Q in the fill: the floor removed, the ordering holds (trees 0.694 on R, 0.637 on Q). ✗● **f**, The fitness side: a ridge over the lower-order terms predicts a held-out triple's fitness at 0.927 on R and 0.734 on Q (test), and CGT's fitness head reads 0.949 and 0.722 on the same triples (validation). ✗● **g**, Measured against predicted score for the boosted trees on arm R test, the predictions cut at their deciles: the median measured score rises with the prediction and the spread within a decile is the part no rung explains. ✗● **h**, Pearson inside bands of the measured score's magnitude on arm R test, bands at its quantiles: every method reads the strong interactions better than the bulk near zero, and the trees lead in every band. ✗● **i**,

between the rungs' predictions on arm R test, Pearson between every pair and with the measured score: the ridge over terms is close to the controls null, the trees add to it (+0.165 paired gain), and the identity ridge agrees with none of them. ✗●

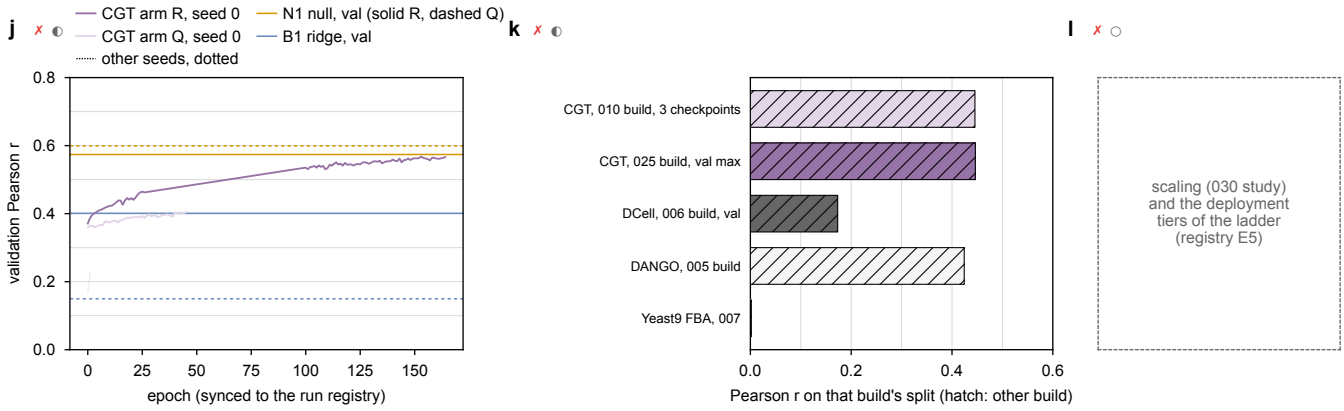


Figure 3: The second sheet: the model's curves and the comparators on other builds. j, CGT's validation interaction Pearson by epoch on both arms, as synced to the run registry (the server holds epochs 0 to 26 and 100 to 164 of arm R, and 0 to 45 of arm Q), with the validation values of the controls null (orange) and the additive ridge (blue) as lines, solid for R and dashed for Q. **k**, The comparators of the earlier draft, each on another build and split, hatched for that reason: the three 010 checkpoints' test mean (0.446), the 025 replication's validation maximum (0.446), DCell's best validation checkpoint on the 006 build (0.173), DANGO on the 005 build (0.424) and Yeast9 FBA (0.0019). **l**, The scaling readout of `030-scaling-study` and the deployment tiers of the ladder (registry E5), not drawn. Written by `figure2_gate_sheet.py`.

The supplement. Three sheets are proposed, none drawn yet. First, how the query decides the number: the same model family on the 010 build (triples only), the 025 S3 closure pool (merged labels) and the 030 per-entry pool (every entry, a dataset token), one bar per build per arm, with the gene-identity ladder of `025-additive-baselines` beside it; what is on file is one seed per build and the ladders. ◗ Second, the screen-offset floor: per-screen and within-screen Pearson of every rung and the transformer, the per-screen mean τ that makes the floor, and the Kuzmin 2018 against 2020 difference in what the arithmetic explains (the controls null reads 0.221 on 2018 rows and 0.683 on 2020 rows of arm R test). ● Third, the comparators of the earlier draft on the same data: DCell, DANGO and Yeast9 FBA refit or rescored on the 030 pinned triples of both arms. ❍

3 The models and comparators on file ✗

The transformer. The cell graph transformer with the perturbed class token read through a linear probe, graph regularization (the KL form of 010) toward nine gene graphs, the full-gene composite sequence embedding (fungal upstream and downstream transformers, CaLM, ProtT5) through a parameter-matched preprocessor, a fitness head beside the interaction head, and the per-entry dataset token of experiment 030, which lets one genotype carry every screen's entry as its own row. Seed 0 on arm R has synced 92 epochs to the registry up to epoch 164, the maximum over epochs is 0.568 at epoch 153 (a biased order statistic), and the last-ten-epoch mean is 0.562; arm Q has synced 45 epochs and reads 0.399 over epochs 36 to 45. Runs: `u28pgzde`, `xy3xnpau`, `xy3xnpau-cont2421912` and `53qqijnu`, `rkwyav5i`. Every seed that carries an arm's tag is read on its own and never stitched into another seed's history; the headline number is seed 0, and a seed with fewer synced epochs than the ten-epoch window is on file as partial and drawn nowhere. Seeds on file, arm R: seed 0, 92 epochs to 164; arm Q: seed 0, 45 epochs to 45; seed 1, 2 epochs to 1 (partial, under the window).

The gene-identity ladder of 025, same arms. Test Pearson on the 025 build, from `025-additive-baselines`: train mean 0.000 and 0.000, screen mean 0.390 and 0.142, query pair only 0.390 and 0.000, additive gene ridge 0.400 and 0.185, additive plus pair 0.406 and 0.180, and the gene-embedding MLP over three seeds 0.432 and 0.141. The 030 ridge of Figure 2 reproduces the 025 ridge on the same arms (0.401 against 0.400; 0.185 against 0.185), which is the end-to-end check that the two builds and the pinned splits agree.

Comparators, each on another build. The earlier draft of Figure 2 placed the transformer beside DCell, DANGO and flux balance analysis. Their results are on file, but on other builds and other splits, so none is a row of the ladder: the three 010 checkpoints read 0.438 to 0.455 on the 010 build's test split and the 025 replication of that configuration reached 0.446 at validation epoch 14; DCell's best checkpoint reads 0.173 (validation, 006 build, one 14.0-day run); DANGO reads 0.424 at best (STRING 11.0, LinearUntilUniform, mean of 2) on the 005 Kuzmin 2018 build; Yeast9 FBA with nothing fitted reads 0.0019 against the released score over 332,313 triples, with 99.9 % of its predictions at zero. Table 1 lists them with their build and statistic. Refitting the three on the 030 pinned triples is entry E7 of the registry.

Table 1: Comparators on file, none on the 030 build. Pearson against the trigenic score; the statistic column says what each number is. Rows written by `figure2_gate_numbers.py`.

model	build, split	split or scope	r	statistic
CGT 010 checkpoint M01	010 build, arm R	test	0.443	one checkpoint
CGT 010 checkpoint M02	010 build, arm R	test	0.438	one checkpoint
CGT 010 checkpoint M03	010 build, arm R	test	0.455	one checkpoint
CGT 025 replication, job 1598	025 build, arm R	val, best epoch 14	0.446	max over epochs, no test
DCell, three checkpoints	006 build	val	0.173	best of 3, one 14-day run
DANGO, STRING 11.0, LinearUntilUniform	005 build, Kuzmin 2018	test	0.424	mean of 2
Yeast9 FBA, nothing fitted	007 build	all 332,313 triples	0.0019	99.9% of predictions at zero

4 What is established and what remains ✗

Established. The published trigenic score is the identity of equation 1 to the rounding floor on both screens, so a held-out triple's null expectation is a function of training records (`025-s3-closure`). The free arithmetic is measured on both arms and is large: 0.577 and 0.566 with no parameters, 0.743 and 0.700 with boosted trees. The transformer as trained reads below the zero-parameter null on both arms (0.562, 0.399, validation, one seed). The gene-identity ridge sits at 0.401 and 0.185, so the transformer does beat the ladder the earlier draft compared it to, and that comparison is the wrong one. The fitness of a held-out triple is predictable from the lower-order terms at 0.927 on R.

Not established. Whether a transformer that reads the lower-order terms as inputs beats the boosted trees, which is the only comparison in which the two regimes match (E1). Whether the transformer's residual over the digenic-controls null carries anything, read as a target (E2). Any test score of a 030 checkpoint, and any seed spread; every transformer number here is one seed on validation (E3, E4). What the ladder reads at deployment tiers, where a new triple has its singles but not its doubles (E5). The comparators on the same data (E7). Why the arithmetic explains far less on the Kuzmin 2018 rows than on the 2020 rows (0.221 against 0.683 on arm R test); hypothesis, untested: the 2020 diagnostic-array screen measured its control queries on the same strains in the same batch as the triples, so the control scores carry the batch, and the 2018 genome-wide screen did not.

Corrections recorded. The 025 document's decision table compared the transformer to the additive ridge and found it below that ridge on arm Q test for one checkpoint; that reading stands, and this document adds that the ridge itself is far under the lower-order null. The weekly note of 2026-10-11 read the Q transformer at 0.391 over epochs 19 to 28; the registry has since synced to epoch 45 and the window moved to 0.399; both are one seed, validation.

5 The decision rules, fixed now ✗

What a panel needs to enter the paper. Every drawn value in a committed result file (●); every transformer bar a test score of a checkpoint selected on validation, over at least three seeds, with the across-seed spread drawn; the regime of every bar named in the panel or its key; the screen-offset floor drawn wherever a pooled Pearson is drawn; and the claim the panel carries written in one sentence in this document before the panel is composed.

The comparison that decides the figure's third claim. The transformer reading the lower-order terms as inputs (E1) against the boosted trees (L2), both on the same pinned test triples, paired by triple over 2,000 bootstrap resamples, three seeds of the transformer. The transformer helps when its seed-mean test Pearson exceeds L2's by at least 0.02 on an arm with the paired interval excluding zero on every seed, and the claim is made per arm. If it does not clear that on arm R, the figure states the lower-order ladder as the measured ceiling of the closure and the transformer as a model of gene identity that reaches part of it, and the manuscript claims no more. If it clears it on R and not on Q, the claim is limited to seen query pairs.

The regime question, settled by naming it. A lookup over measured terms of the same strains is transductive over the closure; a model reading identity is not. The sheets carry the regime in the key and the performance table in a column. A reviewer who objects that the lookup is not a model is answered by E1, which gives the transformer the same inputs; a reviewer who objects that the transformer was handed the answer through the fitness head is answered by Figure 2, which shows it did not take it.

Order of work. E3 first, since it is CPU scoring of checkpoints that exist and turns every validation bar into a test bar. E1 and E4 next, one chain on IGB mmli, the same launcher as the running arms. E5 is CPU and runs beside them. E2, E6 and E7 follow once E1 has read. Nothing is submitted until the author says so.

6 The experiment registry ✗

Shared protocol: the 030 S3 pool (arm R) or S3Q pool (arm Q); pinned validation and test triples never in a loss; the checkpoint chosen on validation interaction Pearson; test Pearson and Spearman, pooled and within screen; three seeds unless stated; paired by triple against the named control over 2,000 bootstrap resamples. Every entry is ❍ until its result file exists.

Table 2: The pre-registered experiments of this gate. E1 and E3 decide the figure's third claim; the rest fill the remaining panels.

id	experiment	
E1	**The transformer with the lower-order terms as inputs.** The 030 composite arm with the thirteen terms of the ladder (and their missing indicators) read by a zero-initialized affine on the interaction logit, and in a second form as a per-triple input token; control: the same arm without them. Both arms, three seeds, 60 epochs on IGB mmli. Decides: whether a model given the same inputs as the boosted trees beats them (the rule of 5).	❍
E2	**The residual target.** The 030 composite arm trained on $\tau + \varepsilon_{ik} + \varepsilon_{jk}$, the part of the score the digenic-controls null does not explain; control: the arm on τ. Arm R, three seeds. Decides: whether the model's signal is in the residual or in the controls it re-learns.	❍
E3	**Test scores of the 030 checkpoints.** CPU scoring of the best-validation checkpoints of both arms on their pinned test triples, as `score_cgt_checkpoint_cpu.py` did for 025, with the per-triple predictions of both heads written beside the ladder's, a paired bootstrap against every rung, and the two derived rungs of 1: C1, the model's f_{ijk} minus the measured null; C2, τ from the model's predicted fitness of every term. Decides: every validation bar becomes a test bar, and whether CGT as a fitness predictor reaches the trees.	❍
E4	**Seeds.** Seeds 1 and 2 of the 030 composite arm on both arms, the budget of seed 0. Decides: the across-seed spread every transformer bar must carry.	❍
E5	**Deployment tiers of the ladder.** The lower-order rungs refit with the terms a new triple would have: singles only; singles and Costanzo doubles where measured; singles and all doubles. CPU, both arms. Decides: the supplement's query panel, and what a strain designer can expect without the Kuzmin doubles.	❍
E6	**Within-screen and Spearman logging.** The trainer logs per-screen and within-screen validation Pearson and Spearman beside the headline, so the floor is visible in every run and the model can be read in rank. Decides: nothing by itself; required by the entry rule.	❍
E7	**Comparators on the same data.** DCell, DANGO and the Yeast9 FBA score refit or rescored on the 030 pinned triples of both arms under the shared protocol. Decides: the supplement's comparator sheet and whether any of them enters the main figure.	❍
E8	**The essentiality holdout arms of 030.** `fit_000` and `embfit_001`, which exclude 698 essentiality singles and serve them as a second validation loader: the AUROC read that links this figure to the essentiality gate. Decides: whether the solid-growth model carries essentiality without being trained on the held-out singles.	❍

7 Every rung, with its regime ✗

Every rung of the lower-order ladder on both arms: validation and test Pearson, the test Pearson within screen (both series centered by screen, which removes the floor), the test Spearman, and the fraction of test rows the rung's own term covered (a missing term falls back to the screen mean for that row). The fitness rung scores f_{ijk}, not τ, and is not on the same scale as the others. The transformer is not a row: it has one seed and no test score, and its validation reads are in 3.

Table 3: The lower-order ladder on arms R and Q, test unless the column says validation. Rows written by `figure2_gate_numbers.py`.

arm	rung	regime	val r	test r	within-screen	Spearman	coverage
R	N0 screen mean	screen only	0.368	0.368	0.000	0.442	1.00
R	N1 digenic controls, zero parameters	lower-order terms	0.574	0.577	0.550	0.630	1.00
R	N2 singles product, zero parameters	lower-order terms	0.267	0.273	0.245	0.304	0.97
R	N1s strain-matched controls	lower-order terms	0.577	0.580	0.551	0.634	1.00
R	B1 additive gene ridge	identity	0.401	0.401	0.177	0.406	1.00
R	L1 ridge, lower-order terms	lower-order terms	0.606	0.609	0.549	0.668	1.00
R	L2 boosted trees, lower-order terms	lower-order terms	0.740	0.743	0.694	0.755	1.00
R	L3 ridge, genes + lower-order terms	both	0.649	0.653	0.581	0.703	1.00
R	F1 ridge, lower-order terms, target f_ijk	lower-order terms	0.927	0.927	0.924	0.929	1.00
Q	N0 screen mean	screen only	0.351	0.378	0.000	0.439	1.00
Q	N1 digenic controls, zero parameters	lower-order terms	0.599	0.566	0.538	0.624	1.00
Q	N2 singles product, zero parameters	lower-order terms	0.351	0.378	0.000	0.439	0.00
Q	N1s strain-matched controls	lower-order terms	0.603	0.573	0.544	0.629	1.00
Q	B1 additive gene ridge	identity	0.150	0.185	0.108	0.164	1.00
Q	L1 ridge, lower-order terms	lower-order terms	0.606	0.584	0.535	0.638	1.00
Q	L2 boosted trees, lower-order terms	lower-order terms	0.668	0.700	0.637	0.718	1.00
Q	L3 ridge, genes + lower-order terms	both	0.622	0.611	0.536	0.658	1.00
Q	F1 ridge, lower-order terms, target f_ijk	lower-order terms	0.738	0.734	0.714	0.698	1.00